

Use of PETase for PET Degradation in a Bioreactor to Improve Effectivity of Recycling

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Abstract

Bacteria with an adaptation to degrade plastic could aid in the recycling process for plastic to become less detrimental to the environment. As industries continue to mass produce plastic, it is becoming more and more apparent that there is no effective way to recycle it and address this pollution. The goal of this project is to engineer a device to effectively degrade plastic. Nine billion tons of non-biodegradable plastic waste pollutes the earth. While recycling is a good way to address this, manufacturers do not reuse plastic as freshly producing it costs less than recycled products. Some organisms have evolved from exposure to plastic and developed enzymes to break down plastic for energy. One common type of plastic polymer is polyurethane terephthalate (PET). *E. coli* K12 was cultured in a liquid medium with PET film in a glass jar. The pH and carbon dioxide levels were recorded over a span of 168 hours at 30 degrees Celsius. The mass of the PET film was recorded before and after and found to decrease as it was dissolved. Because CO₂ levels increased, cellular respiration was performed using carbon from the PET, which in turn made the solution more acidic. With this adaptation, these bacteria could be used to dissolve plastic and cultured as needed to supply an industrial demand, overall creating a more effective system of recycling. In the future, isolating the products from this reaction could be investigated further to then use them to remake plastic, resulting in a cyclical process.

Keywords: bacteria, bioreactor, enzyme, recycling, PET

Acknowledgements

Thank you to my mother for funding the purchases needed for this experiment, as well as my teacher Dr. Crowthers for guiding me in this project.

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Use of PETase for PET Degradation in a Bioreactor to Improve Effectivity of Recycling

As industries continue to mass produce plastic, it is becoming more and more apparent that there is no effective way to recycle plastic and address plastic pollution. In fact, by 2050 it is expected that plastic production will grow by 70% to nearly 600 million tons per year (Cornwall, 2021). Many of these plastic products are made from non-biodegradable plastic polymers, and paired with the unsustainable disposal habits and lack of functional waste management systems, this plastic ends up polluting the earth.

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Pollution & Recycling

About 9 billion tons of plastic waste pollutes the earth (Demircan & Keskin, 2020). A majority of this was produced within the past two decades, evidence of the increased production of plastic in recent years to meet the higher demand due to its versatility. However, with this versatility comes a drawback, as these plastics do not easily decay in nature (Demircan & Keskin, 2020).

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Recycling is one process that aims to reutilize wastes that would otherwise add to environmental pollution. While recycling is a good way to address plastic pollution, most manufacturers choose not to reuse plastic as producing new plastics costs less than producing plastic from recycled products. Only 9% of all recyclable plastics end up being recycled in the US, since at every step of the process more and more of the recyclable material is disregarded, or it ends up in landfills and ultimately incorrectly disposed of (Sheth et al., 2019).

Other ways to remove plastic waste is to burn it, but this releases pollutant gases such as sulfur dioxide, hydrogen cyanide, and hydrogen fluoride, which have been shown to be toxic and poisonous (Bellini, 2019). Burying it is another option, but this pollutes the land and water,

and mechanically breaking it down is expensive (Demircan & Keskin, 2020). Alternatives to common uses of plastic, such as aluminum or glass cans, release more carbon into the environment than plastic, making them less beneficial alternatives (Stanton et al., 2020). Even when the material changes, humans will continue to litter, making it essential to find a way to actively recycle and properly dispose of wastes (Stanton et al., 2020).

Current Functionality of PETase & Limitations

Some organisms have evolved mechanisms that break down plastic for energy due to exposure to plastic (Hiraga et al., 2019). Microorganisms found in areas containing trash and vegetation, namely plastic landfills, have adapted to contain enzymes that can decompose plastic polymers, including the polymers that are used in bottles and clothing (Cornwall, 2021). One organism, *Ideonella sakaiensis* 201-F6, was found in a Japanese landfill that had enzymes to degrade PET. These enzymes were known as PETases, which can dissolve PET by using it as a source of energy and carbon and ultimately breaking PET down into its monomers (Hiraga et al., 2019).

Ester bonds that link PET monomers can be hydrolyzed using enzymes, allowing the plastic to be biodegraded by breaking the carbon-carbon bonds (Hiraga et al., 2019). These carbon-carbon bonds are what make the polymer more difficult to degrade when compared to other plastics such as polyesters that have carbon-oxygen bonds (Cornwall, 2021).

The bacteria latch on to the PET and form a biofilm - a colony of bacteria that rely on the material for nutrients (O'Toole et al., 2000). However, some bacteria contain enzymes that have been deemed to not work fast enough and others are only able to break down a specific plastic polymer, and these qualities are undesirable for the recycling and manufacturing process as it is

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harder to implement this technology when it takes a long time for the process to be completed (Cornwall, 2021). These limitations have prompted studies to improve the enzyme. While many experiments have looked at ways to improve the efficiency of PETase, little progress has been made to utilize this technology to address the growing issue of plastic pollution and ineffective recycling. Additionally, minimal research has been done to try and culture bacteria with this enzyme without exposure from a landfill or what conditions may be needed for this to be possible.

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Problem Statement

As industries continue to mass produce plastic, it is becoming more and more apparent that there is no effective way to recycle plastic and address plastic pollution.

Objective

Obj. 1a: The goal of this project was to engineer a device to effectively degrade plastic.

Obj. 1b: The goal of this project was to determine the conditions needed to dissolve plastic in a bacterial solution.

Hypothesis

Hyp. 1a: It is hypothesized that E. coli K12 cultured in an environment with a low carbon media and PET will result in a decreased mass of the PET.

Hyp. 1b: It is hypothesized that E. coli K12 cultured in an environment with a low carbon media and PET will result in an increased level of carbon dioxide.

Hyp. 1c: It is hypothesized that E. coli K12 cultured in an environment with a low carbon media and PET will result in an increased pH.

Section II: Methodology

Role of Student vs. Mentor

Over the course of 6 months, I was responsible for researching, planning, and executing this project. My teacher, Dr. Crowthers, assisted me with the proper technique and methods to conduct this experiment and sub-cultured the bacteria that was used in the assay.

Equipment and Materials

The procedure by Odobašić (2020) was followed to make the YSV media and culture the bacteria. 200 mL of the media was placed in a mason jar, and plastic filament was sterilized with ethanol and placed in the jar with a stir bar. An incubator was used to culture the bacteria, E. coli K12, for the duration of the experiment, and the stir plate was placed in the incubator with the four jars on it and taped shut.

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Measuring pH

A pH meter was sterilized with bleach after each use to measure the pH of the solution to see if the carbon from the plastic filament was being used.

Measuring CO₂

A CO₂ sensor was used by placing it through a hole in the top film to observe if the carbon from the plastic filament was being used. Once it reached a steady reading, this number was recorded, and the jar was recovered and placed back in the incubator.

Measuring Mass

The mass of the plastic was measured before and after the experiment occurred so as to not disturb the formation of a biofilm on the plastic in order to measure if the plastic had been significantly degraded.

Statistical Tests

A Chi Square test was used for the carbon dioxide levels and pH with the initial values as expected values in order to see if there was a significant difference between the control results and experimental results. Additionally, a Chi Square test was used to compare the initial and final masses.

Chi Square test

Chi Square tests were used for each group to compare the initial and final pHs, carbon dioxide levels, and masses to determine if a significant change occurred during the incubation period.

Section III: Results

A culture system was set up to monitor the degradation of PET over a span of time and compared to the degradation of ABS, PLA, and a group without a filament. Changes were observed in regard to the pH, carbon dioxide levels, and mass of the different filament groups in this experiment.

pH Results

It was found that the pH in all groups went up and then began to decrease. The pH of PET was 8.31 112 hours in, and at 168 hours it had decreased to 7.55, however, a similar trend was seen in all groups as detailed in Figure 1. From the Chi Square tests, p values of 0.782,



Figure 1: pH graphs of the acidity at different time intervals for the PET, PLA, ABS, and no plastic filament groups.

0.718, and 0.655 were found respectively, signifying that there was not a significant change from the base pH in either group.

CO₂ Results

It was found that the carbon dioxide levels increased for PET and PLA filament groups by an average of 6 ppm, but not for the ABS filament group over the span of 168 hours as seen in Figure 2. Based on the p values of 0.91 from the Chi Square tests, there was not a significant change in carbon dioxide levels over the span of the experiment.

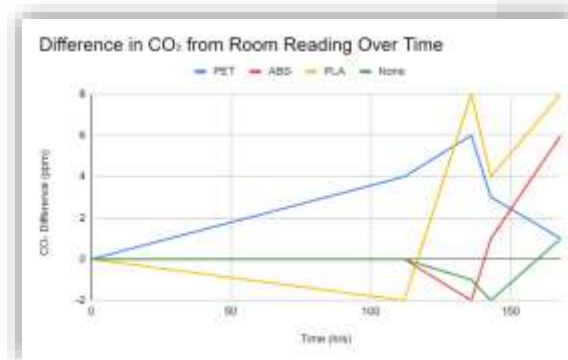


Figure 2: Carbon dioxide graphs of the level over the time interval for the PET, PLA, ABS, and no plastic filament groups.

Mass Results

When comparing the masses before and after the testing, the mass for PET increased by 0.02 grams, while the mass for PLA increased by 0.0075g and the mass for ABS decreased by 0.0167g. Based on the Chi Square test, these changes in mass were not significant, resulting in p-values of 0.983, 0.985, and 0.994 respectively from the Chi Square tests.

Section IV: Discussion

The goal of creating a device that used bacteria to degrade plastic was not achieved at this time. For the first hypothesis that the mass would decrease, it was found that this was not supported as the mass was not statistically different ($p > 0.1$, students t-test) in comparison to the control. However, since the mass was taken in a way so as to not disrupt the biofilm, this was also measuring the mass of the bacteria, meaning that the increase in mass, although not

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significant, was likely the bacteria that were adapting to use the plastic as a carbon source. For the second hypothesis that the carbon dioxide would increase, was also not significantly supported, with p values of 0.91 for all three groups. For the last hypothesis that the pH would decrease, this was not supported based on the p values of 0.78, 0.72, and 0.66 for PET, ABS, and PLA respectively.

While this experiment showed no statistically significant results, it is an indication that selecting bacteria from a random sample and culturing them to develop a reliance on plastic for carbon may not be a feasible way of producing bacteria with this enzyme for industrial use in a device. Certain bacteria such as *I. sakaiensis* may be needed as a base organism to degrade plastic, as mentioned earlier in the processes of published studies such as Hiraga et al. (2019).

Limitations

One limitation was the accuracy of data due to the testing environment. Since the testing location was in a classroom, the carbon dioxide level in the room was higher at the end of the day by about 200ppm, skewing the readings of the jars since they were not set up in a completely closed environment. To address this, carbon dioxide levels were analyzed as the difference between the room level and the observed reading. Additionally, because the testing period was limited, the recorded results were not as extensive as expected.

Statistical Analysis

Chi Square tests were used to compare the initial results to the final results after a period of time to see if there was any statistically significant change in the readings. It was

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found that there were no significant differences between the initial and final readings for the mass, carbon dioxide, and pH measurements.

While many past studies, such as Demircan, T. & Keskin, E. (2020), looked at culturing *Ideonella sakaiensis*, this experiment attempted to culture bacteria that could carry out the same process as *I. sakaiensis*. Because of this, there would be a greater ability to culture these bacteria as needed to supply the industry with a device to degrade plastic.

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Application

While no data was shown to support the usage of bacteria to degrade plastic, this experiment provided a beginning to work towards creating a process that may work. By beginning this process, changes can be made to refine the procedure and testing environment to continue to work towards creating a more sustainable recycling system.

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Future Research

In the future, more research could be done to optimize the media for the growth of plastic-degrading bacteria. This would allow for more ideal growth of bacteria, and the creation of a cyclic process for recycling so that less plastic ends up in landfills and polluting the earth. Additionally, the next step would be to develop a process to separate the ethylene glycol and terephthalic acid from the rest of the solution and media to be able to reuse these substances for plastic.

Section V: Conclusion

This project aimed to create a device to degrade plastic, and this would be evident based on an increase in carbon dioxide and acidity, along with a decrease in mass. To do this, 4 jars were placed on a stir plate in an incubator with minimal carbon media, a stir bar, and PET, PLA, or ABS. Measurements of carbon dioxide, pH, and mass were taken and compared to see if any significant changes occurred over the period of a week. It was found that there were no significant changes in any group based on the p-values consistently above 0.5 for each group. The pH of PET was 8.31 112 hours in, and at 168 hours it had decreased to 7.55, however, this general trend was seen across all plastic groups, including the group without a filament. It was found that the carbon dioxide levels increased for PET and PLA filament groups by an average of 6 ppm, but not for the ABS filament. The mass for PET increased by 0.02 grams, while the mass for PLA increased by 0.0075g and the mass for ABS decreased by 0.0167g. The mass likely increased due to the formation of a biofilm, which suggests that given more time the bacteria would have adapted and degraded the plastic further. This research is critical to determining the functioning criteria for similar devices to address the growing issue of plastic pollution and ineffective recycling.

Section VI: References

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Section VII: Appendices

Appendix A: Limitations and Assumptions

Limitations:

Limitations are components of your project/experiment that are considered out of your control. These components are inherently part of your approach. You can provide a list of limitations in your project.

1. Mutated enzymes were not available.
2. Only one carbon sensor was available, so jars were not perfectly sealed.
3. Testing location was in a commonly used area with breathing people.
4. Putting the stir plate in the incubator resulted in the incubator not being able to be shut completely.

Assumptions:

1. The released carbon dioxide stayed in the local area of the jar.
2. The observed results could vary based on bacteria sample.

Use of bioreactor to select PET-degrading bacteria

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